QP CODE : 25901240

MGU-UGP (HONOURS)REGULAR EXAMINATION, MARCH 2026
FOURTH SEMESTER
MG4DSCECC200 – ELECTRONICS SERVICE TECHNOLOGY
(2024 Admission Onwards)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Interest(I), Appreciation(Ap) and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

SCHEME OF EVALUATION

PART A
Multiple Choice Questions
Answer all questions. Each carries 1 mark

		Learning Domain	CO No(s)
1.	(b) Overload	U	CO1
2.	(c)Semiconductor Material	U	CO2
3.	(c) Temperature increases	A	CO3
4.	(d)Wood,plastic and thin metal	U	CO2
5.	(b)Brokem track	U	CO3
6.	(c)Wattmeter	U	CO1
7.	(b)Faster setup and high accuracy	K	CO2
8.	(c)Circuit malfunction	K	CO3
9.	(b)Short Circuit	K	CO1
10.	(b)Electromagnetic Induction	U	CO2

PART B
Short Answer Questions
Answer 4 questions. Each carries 5 mark

11.	Definition: Electric potential difference between two points (1 mark) Unit: Volt (V) (1 mark) Formula: V = W/Q (1 mark) Physical meaning: Work done to move charge (1 mark) Significance: Drives current in a circuit (1 mark	K	CO1
12.	Definition of taper (1 mark) Ensures proper fitting of components (1 mark) Prevents mechanical looseness (1 mark) Important in shafts, connectors, tools (1 mark)	U	CO2

	Improves alignment and performance (1 mark)		
13.	Any five difference-1 mark each	U	CO3
14.	Definition (1 mark) Based on Faraday's law (1 mark) Measures flow of conductive liquids (1 mark) No obstruction in flow (1 mark) Application: Water supply, chemical industries (1 mark)	K	CO2
15.	Shows circuit connections (1 mark) Helps in troubleshooting (1 mark) Identifies components and paths (1 mark) Reduces repair time (1 mark) Ensures safety (1 mark)	A	CO3
16.	Risk of wrong connections (1 mark) Electrical accidents/shock (1 mark) Equipment damage (1 mark) Difficulty in maintenance (1 mark) Confusion in troubleshooting (1 mark)	K	CO1

PART C
Essay Questions
Answer any 2 questions. Each carries 10 mark

17.	Definition (2 marks) Principle: Ultrasonic wave propagation (2 marks) Types: Transit time / Doppler (2 marks) Working explanation (3 marks) Advantages & applications (1 mark)	K	CO2
18.	Definition (2 marks) Calculation method (connected load, demand factor) (3 marks) Importance: Prevents overloading (1 mark) Proper cable selection (1 mark) Safety (1 mark) Energy efficiency (1 mark) Practical example (1 mark)	U	CO1
19.	Identification of terminals (2 marks) Use of diode mode (2 marks) Base-emitter test (2 marks) Base-collector test (2 marks) Fault identification (open/short) (2 marks)	U	CO3
20.	Definition of RCCB (2 marks) Working principle (leakage current detection) (2 marks) Importance in safety (2 marks) MCB definition (1 mark) Comparison (3 marks)	U	CO1